

Smaller reactors for reduced-voltage starting and motor-control-center current-limiting applications are available already designed; therefore, it usually is not economical to custom design smaller reactors solely for energy conservation.

Where the opportunity does exist for energy conservation in the larger reactors, the specifier can advise the manufacturer about the loss evaluation dollars per kilowatt as follows:

- *Shunt reactor.* A shunt reactor acts as a constant load at a given voltage. Its total loss cost evaluation (even though consisting mainly of I^2R losses) is calculated using only the no-load formula. Its losses increase as the impressed voltage increases.
- *Series reactors.* A series reactor experiences varying load. Because it does not have a no-load loss, its total power loss cost evaluation is calculated using only the load loss formula.

5.5 Capacitors and synchronous machines

5.5.1 Quality of power

Energy savings can be achieved by improving the quality of the power supply to the utilization equipment.

The quality of the power supply is defined by five independent parameters:

- a) *Nominal supply quality.* The magnitude, harmonic content, and balance of the supply voltage waveform under normal conditions.
- b) *Long-term reliability.* The frequency and extent of long-term power outages (greater than 10 s) such as caused by a permanent fault.
- c) *Short-term interruption.* The frequency of short-term total supply interruptions (less than 10 s) such as an auto recloser operation.
- d) *Short-term disturbances.* The frequency and magnitude of short-term disturbances (less than 10 s), including voltage drops, such as caused by reflected faults from adjacent feeders.
- e) *Transient disturbances.* The frequency and magnitude of transient disturbances (often less than one cycle duration and no more than several seconds), including impulses, voltage surges (spikes), and voltage dips (sags).

The nominal voltage to the service point may vary according to ANSI C84.1-1989 [B26] within the limits of Range A or 105% to 95% of the nominal system voltage on a regulated power distribution system. On an unregulated power distribution system the voltage may vary within Range B or from 105.8% to 91.7% of the nominal system voltage. Refer to IEEE Std 141-1993 for a further discussion.

Knowing the allowable voltage range at the service, the responsible industrial facility staff can set the transformer taps, or the automatic tap changing control, to provide the optimum voltage within the facility for most efficient running of motors and other equipment.

The system frequency at the supply service may vary within the range of 60 Hz plus or minus 0.1 Hz. This is outside the control of the responsible industrial facility staff unless frequency converters are installed.

The power supply system designed by the utility should be capable of keeping the maximum system unbalance to below 3%.

5.5.1.1 Harmonic limits

5.5.1.1.1 Definitions

distortion factor (harmonic factor): The ratio of the root-mean-square of the harmonic content to the root-mean-square value of the fundamental quantity, expressed as a percent of the fundamental.

$$\text{DF} = \left[\frac{\text{sum of the squares of amplitudes of all harmonics}}{\text{square of amplitude of fundamental}} \right]^{1/2} \cdot 100\%$$

filter: A generic term used to describe those types of equipment whose purpose is to reduce the harmonic current or voltage flowing in or being impressed upon specific parts of an electrical power system, or both.

harmonic: Sinusoidal quantities, the frequencies of which are whole multiples of the 60 Hz fundamental frequency.

harmonic analyzer: An instrument for recording and displaying the waveform of the electrical circuit into its component frequencies and amplitudes.

harmonic factor: The ratio of the root-sum-square (rss) value of all the harmonics to the root-mean-square (rms) value of the fundamental use:

$$\text{harmonic factor (for current)} = \frac{(I_3^2 + I_5^2 + I_7^2)^{1/2}}{I_1}$$

$$\text{harmonic factor (for voltage)} = \frac{(E_3^2 + E_5^2 + E_7^2 + \dots)^{1/2}}{E_1}$$

harmonic order (*n*): The ratio between the harmonic frequency and the fundamental frequency.

nonlinear load: A load that draws a nonsinusoidal current wave when supplied by a sinusoidal voltage source.

point of common coupling (PCC): Point at which the customers supply tees off the general utility system supplying other customers.

total demand distortion (TDD): The total root-sum-square harmonic current distortion in percent of the maximum demand load current (15 min or 30 min demand).

total harmonic distortion (THD): This term has come into common usage to define voltage or current “distortion factor.”

triplen harmonics: Harmonics having a frequency exactly divisible by three times the fundamental frequency.

voltage unbalance: The voltage unbalance, in percent, is 100 times the maximum voltage deviation from the average voltage, divided by the average voltage:

$$\text{Voltage: unbalance (\%)} = 100 \cdot \frac{\text{maximum voltage deviation from average voltage}}{\text{average voltage}}$$

5.5.1.1.2 General

The problem of loads with nonlinear characteristics depends upon their connection within the building's electrical distribution system

There are two sets of standards for harmonic limits as described:

- a) *Utility harmonic limits (harmonic limits for suppliers).* Electric power provided to the building is required to have a quality defined by the voltage distortion content as measured at the *point of common coupling* (PCC).

Bus voltage at PCC	Individual voltage distortion (%)	Total voltage distortion (THD%)
69 000 V and below	3.0	5.0

- b) *Consumer harmonic limits (harmonic limits for buildings/industrial facilities).* Consumers (the building owners/operators) are required to limit the harmonic current injection from their equipment into the utility system as measured at the PCC. The harmonic current limits are based on the size of the power system. The following table shows current distortion limits for general distribution systems (120 V-69 000 V). The maximum harmonic current distortion is in percent of maximum demand load current individual harmonic order (odd harmonic).

$I_{sc}I_L$	$h > 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h$	TDD
< 20	4.0	2.0	1.5	0.6	0.3	5.0
$20 < 50$	7.0	3.5	2.5	1.0	0.5	8.0

where

I_{sc} is the utility maximum short-circuit current at the PCC
 I_L is the maximum demand load current (60 Hz) at the PCC

NOTE — The above table, from table 10.4 of IEEE Std 519-1992 [B127], continues for other power systems. The usual consumers have power systems in the range of I_{sc}/I_L less than 20.

- c) *Equipment exceeding harmonic limits.* When it is found by the measurement survey that the harmonic limits are exceeded, then some or all of the following procedures may be followed to reduce the distortion to acceptable levels:
- 1) Rearrangement of the incoming power supply conductors to a different utility distribution line or to a larger capacity utility distribution line. This may avoid poor power quality caused by another utility customer.
 - 2) Equipment with three-phase electronic *uninterruptible power supply* (UPS) systems, for power conditioning and supply during utility outages, should use 12-pulse rectifiers to minimize distortion on the input feeders and 12-step inverters to eliminate all harmonics below the 11th.
 - 3) Equipment with three-phase adjustable speed drives, for motor speed control, should use harmonic filters to reduce or eliminate the 5th, 7th, and 11th harmonics or others.
 - 4) Run a dedicated neutral conductor with each phase conductor when supplying single-phase loads with harmonic content.
 - 5) Run a full-size or oversized neutral conductor on 4-wire, 3-phase circuits where the major portion of the load consists of nonlinear loads, such as electric-discharge lighting (fluorescent and high-intensity discharge lighting), electronic computers (data processing), or similar equipment, where there are harmonic currents in the neutral conductor.
 - 6) In panelboards, switchboards, and motor control centers where there may be single-phase loads, rearrange the circuits among the phases to balance the currents so that the current in the neutral due to phase unbalance is reduced or eliminated.
 - 7) Use isolation transformers in branch circuits with harmonic distortion problems. For example, a delta-wye isolation transformer may improve the power factor, block neutral currents from transferring further up the line, lower the current distortion on the branch circuit, and act to reground the secondary neutral to lower the neutral-to-ground voltage problems.
 - 8) Use active power line conditioners (APLCs) to cancel harmonics. An active power line conditioner is a compact, integrated power conditioning device that helps to solve a variety of power quality problems. APLCs combine adaptive, active harmonic filtering, effective line voltage regulation, and transient voltage surge suppression to protect sensitive electronic equipment. APLCs are available in single-phase